

Donghwan Chi

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Interests

Multimodal Large Language Models, Vision-Language-Speech Generation, Object-Centric Representation Learning, Text-to-Image Generation, Multimodal Alignment

Education

M.S. Korea University, Artificial Intelligence

March 2025 – Present
(expected Feb. 2027)

- Advisor: Sungwoong Kim
- GPA: 4.5/4.5

B.S. Korea University, Computer Science and Engineering

March 2019 – Feb. 2025

- GPA: 4.11/4.5
- Leave of absence for mandatory military service in the Republic of Korea Air Force, Jan. 2021 – Jan. 2023

Publications

Slot-MLLM: Object-Centric Visual Tokenization for Multimodal LLM

Submitted to IEEE Access.
arXiv:2505.17726

Donghwan Chi^{*}, Hyomin Kim^{*}, Yoonjin Oh, Yongjin Kim, Donghoon Lee, Daejin Jo, Jongmin Kim, Junyeob Baek, Sungjin Ahn, Sungwoong Kim^{*} *Equal contribution.*

Object-centric Self-improving Preference Optimization for Text-to-Image Generation

CVPR 2026 Poster.

Yoonjin Oh, Yongjin Kim, Hyomin Kim, **Donghwan Chi**, Sungwoong Kim

Research Experience

WAVLab, Carnegie Mellon University

Pittsburgh, USA (Online)

Visiting Collaborator (Advisor: Prof. Shinji Watanabe)

Oct. 2025

- Conducted research on vision-speech-language omni-modal models for unified multi-modal understanding and generation.
- Developed alignment-preserving training methods to extend large vision-language models with discrete speech tokenizers for ITS2ITS omni-modal understanding and generation across image, text, and speech.

Present

AGI Lab, Korea University

Founding Member & Graduate Researcher (Advisor: Prof. Sungwoong Kim and Prof. Sunghin Lim)

Seoul, South Korea

Sept. 2023

Present

- Led research on object-centric visual tokenization for multimodal large language models, resulting in Slot-MLLM.
- Designed a Slot Q-Former-based visual tokenizer to improve object-level grounding, generation, and editing.
- Conducted large-scale multimodal training on image-text and interleaved datasets using multi-GPU distributed training.
- Evaluated object-centric reasoning and generation on benchmarks including spatial relation understanding, text-to-image generation, and image editing.
- Built and administered lab compute infrastructure from inception (4 multi-GPU servers, 20 researchers); managed lab GitHub organization.
- Led multi-node distributed training on national GPU clusters (IITP)

Extracurricular Activities

Artificial Intelligence Korea University (AIKU),

First Cohort Senior Member, Technical & Academic Management Team

Seoul, South Korea

Jan. 2023 – Jan. 2024

1 year

- Managed the club's servers, ensuring reliable and efficient operation.
- Served as a committee member for multiple hackathon question designs.
- Conducted and participated in a project related to diffusion models.
- Organized and taught introductory courses for new members.

Honors and Awards

- **Chief of Naval Operations Republic of Korea Navy Award**, Open Source Academy Military (OSAM) Hackathon, 2021
- **Semester High Honors**, Korea University, 2019, 2020, 2023, 2024
- **Special Scholarship by Student Affairs Office**, Korea University, Fall 2023, Spring 2024
- **Academic Excellence Scholarship**, Korea University, Spring 2024
- **Excellence Award**, Capstone Design Competition, Korea University, Fall 2024

Technologies

Languages: Python, Bash, LaTeX

ML Frameworks: PyTorch, PyTorch Lightning, Hugging Face Transformers

ML Systems: Distributed Training, Multi-GPU Training, Mixed Precision Training, DeepSpeed, Weights & Biases

Research Areas: Multimodal LLMs, Vision-Language Models, Diffusion Models, Speech Tokenization

Language Proficiency

TOEFL iBT: 5.0 (100)